How to use Airline Cancellation Analysis Dashboard?

Navigator

- Guideline
- Operation Overview
- Airline Performance
- Airport Performance
- Route Map

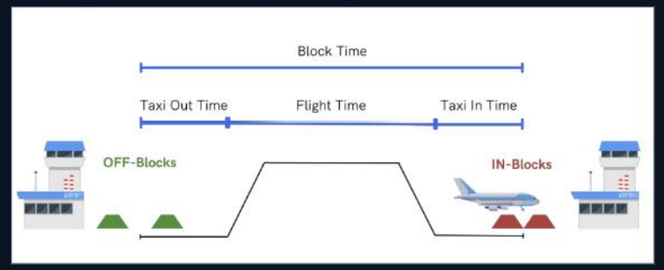
This dashboard analyzes flight operation in 2015. The goal is to identify delay trends, airline & airport performance, and how time & geographic factors affect journeys to improve performance.

Interactions:

- Hover → see tooltips with trends and breakdowns
- Right-click airline → Drill through → Details
- Click & choose an airline name or use filters (Month, Airline, Airport) to narrow the view

KPI Glossary:

- A15 On-time Arrival Flights/ D15 On-time Departure: a flight is "on-time" if it arrives or departs within 15 minutes after schedule. More than 15 minutes is delayed
- Taxi In/ Taxi Out time: see the image



Delay reasons:

- Airline-caused delay = Airline Delay + Late Aircraft Delay
  - Airline Delay: delay by airline-controlled issues, e.g., maintenance, unavailable crew, late boarding due to airline ops etc.
  - Late Aircraft Delay: delay by inbound aircraft arrived late. For example, aircraft of previous flight arrived late, so it couldn't depart on time for this flight. It's basically a domino effect in the airline network.
- Airport-caused delay = Air System Delay + Security Delay
  - Air System Delay: delay by ATC system. For example, ATC holds departures because airspace is crowded
  - Security Delay: delay by security-related reasons. For example, security screening incident slows down boarding or airport operations
- Weather delay: delay by weather impact, e.g., snowstorm, thunderstorm, low visibility etc.

Overview

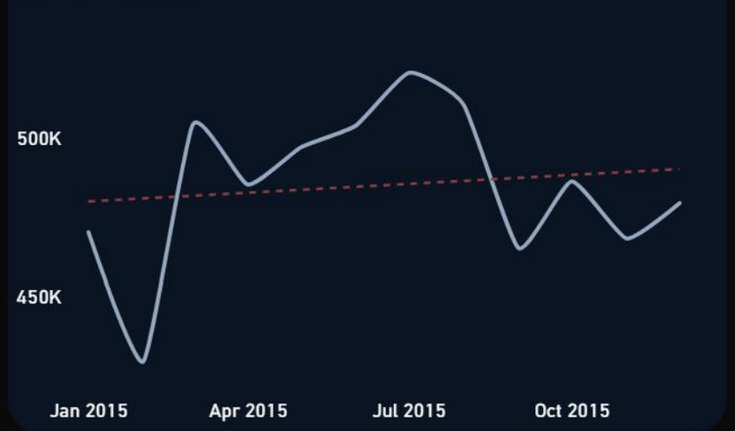
Total Flights	A15 Arrival Flights	D15 Departure Flights	Avg Arrival Delay (min)	Cancellation Rate	Diversion Rate
5.82M LM: n/a	4.69M LM: n/a	4.71M LM: n/a	61 LM: n/a	1.54 % LM: n/a	0.26 % LM: n/a

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Total Flights by Month

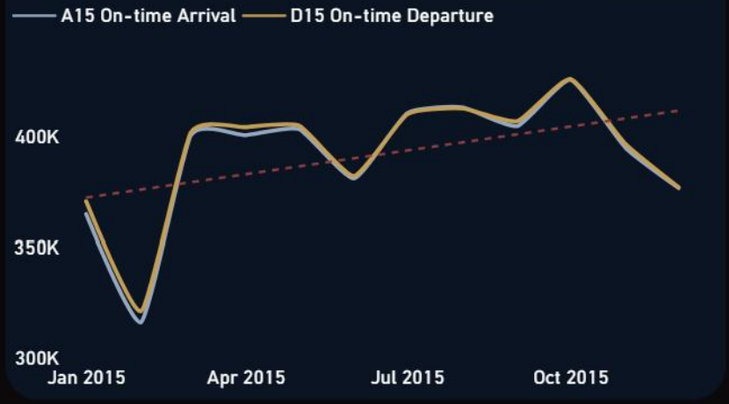
Hover for more information



Total A15 On-time Arrival & D15 On-time Departure Flights by Time

Switch to view

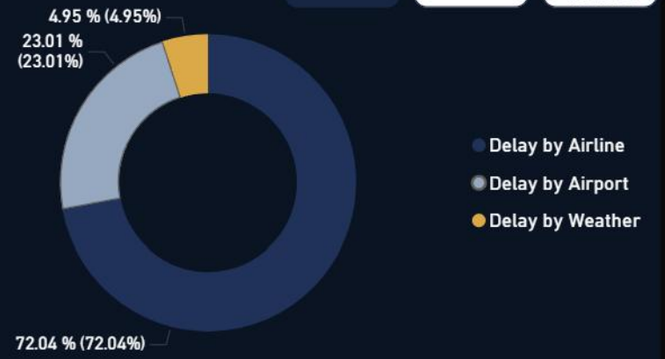
Total Flights | Rate



Total Arrival Delay (min) by Cause

Switch to view

Overall | By Airline | By Airport & Weather

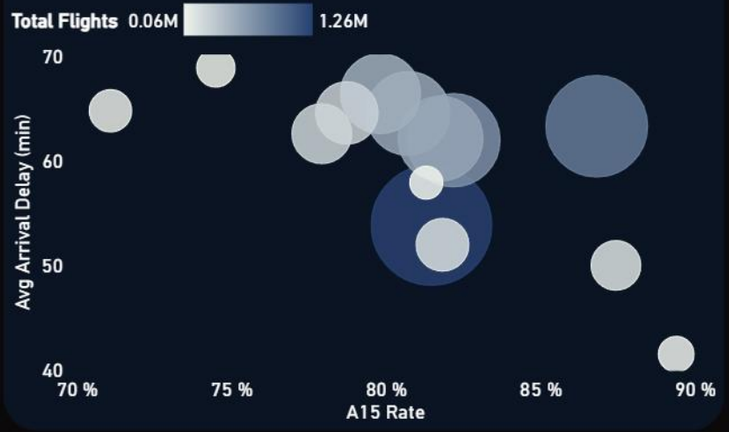


Filter

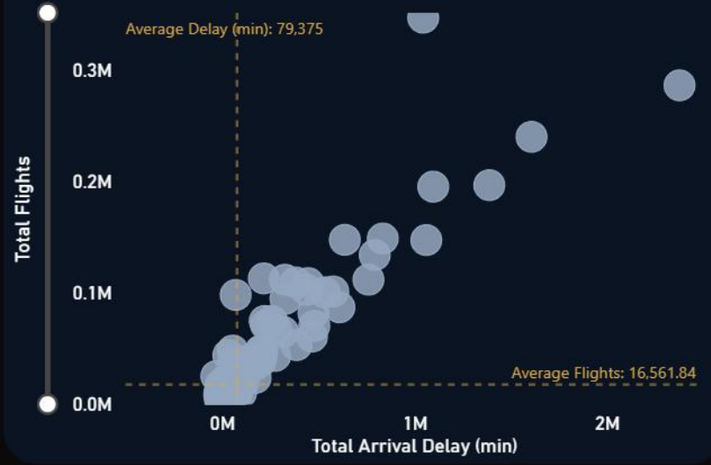
- Month Name: All
- AIRPORT: All
- AIRLINE: All

A15 & Avg Arrival Delay Rate by Airline

Hover for more information

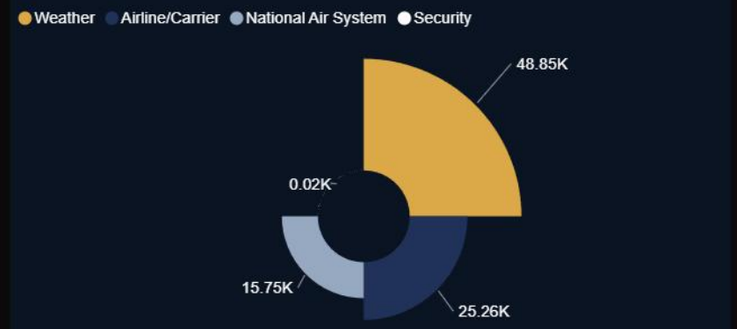


Total Flights & Total Arrival Delay by Airport



Total Cancelled Flights by Reasons

Hover for more information



Airline Performance

Total Flights	A15 Arrival Flights	D15 Departure Flights	Avg Arrival Delay (min)	Cancellation Rate	Diversion Rate
5.82M LM: n/a	4.69M LM: n/a	4.71M LM: n/a	61 LM: n/a	1.54 % LM: n/a	0.26 % LM: n/a

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Right-click an airline and select *Drill through → Drill through Airline* for more information

Arrival Delay Rate by Airline

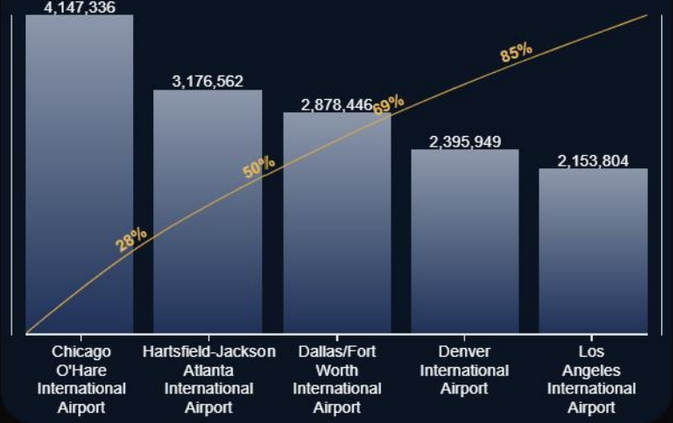
Total A15 | Delay Rate

Hover for more information



Top 5 Airports Causing Arrival Delay (min)

Hover for more information



Total Cancelled Flights by Airline & Reason

Hover for more information



Filter

- Month Name: All
- AIRPORT: All
- AIRLINE: All

Total Arrival Delay (min) Caused by Airport & Weather

Hover for more information

Air System | Weather | Security



Total Arrival Delay (min) Caused by Airline

Hover for more information

Late Aircraft | Airline Delay



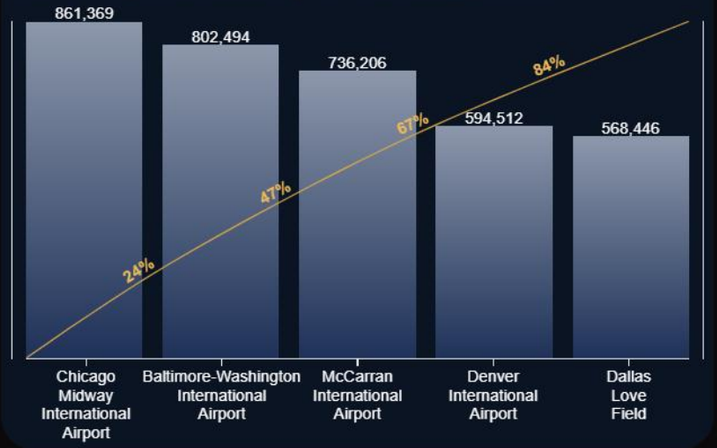
Average Arrival Delay (min) by time of a day

AIRLINE	00–04	05–07	08–10	11–13	14–16	17–19	20–23
Alaska Airlines Inc.	41	48	51	51	50	50	50
American Airlines Inc.	60	61	59	61	62	65	61
American Eagle Airlines Inc.	60	93	69	62	63	63	62
Atlantic Southeast Airlines	53	81	65	63	64	66	63
Delta Air Lines Inc.	57	64	65	61	62	65	64
Frontier Airlines Inc.	58	69	72	71	70	69	68
Hawaiian Airlines Inc.		50	43	45	40	37	41
JetBlue Airways	62	54	51	55	63	66	67

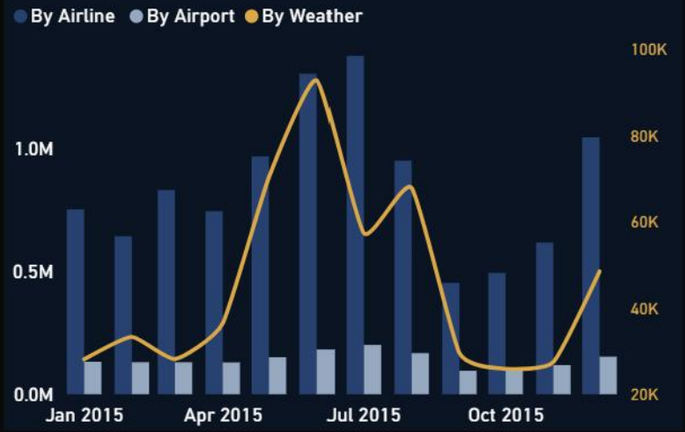
Airline In Details | Southwest Airlines C...

Total Flight	A15 Flight	Avg Arrival Delay (min)	Cancelled Flights	Diverted Flights	Routes
1.26M	1.02M	54	16K	3K	3K

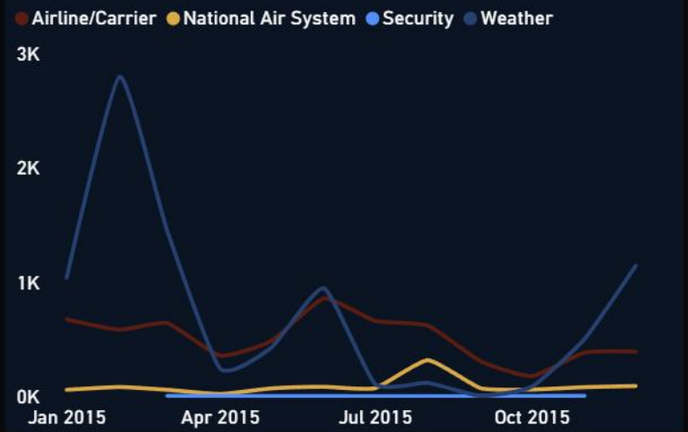
Top 5 Airports Causing Arrival Delay (min)



Delay (min) by Month



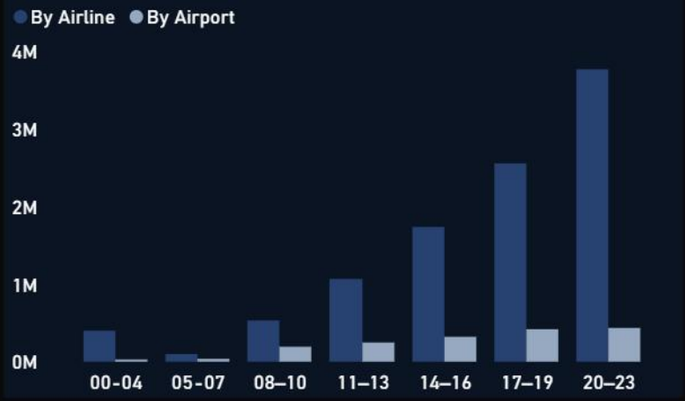
Cancellation Flights & Reasons by Month



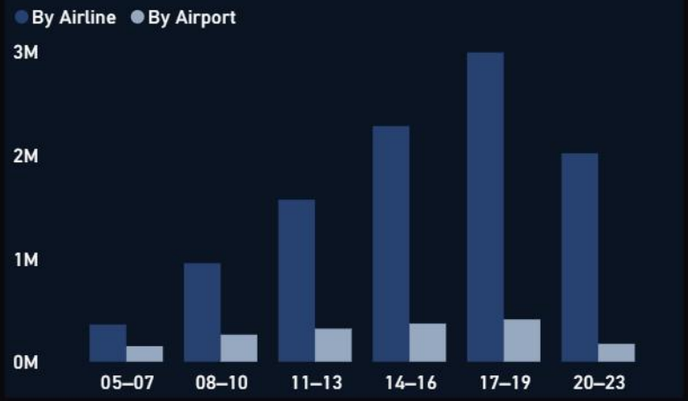
Top 5 Cities Causing Worst Arrival Delay



Reasons Causing Delay (min) by Arrival Time Bucket of a day



Reasons Causing Delay (min) by Departure Time Bucket of a day



Airport Performance

Total Flights	A15 Arrival Flights	D15 Departure Flights	Avg Arrival Delay (min)	Avg Taxi In (min)	Avg Taxi Out (min)
5.82M	4.69M	4.71M	61	7.43	16.07
LM: n/a	LM: n/a	LM: n/a	LM: n/a	LM: n/a	LM: n/a

Navigator

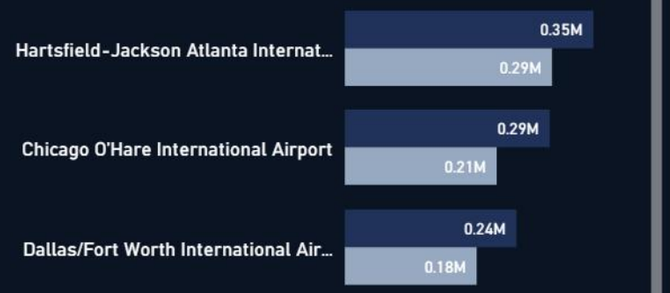
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Total Flights & A15 Flights by Airport

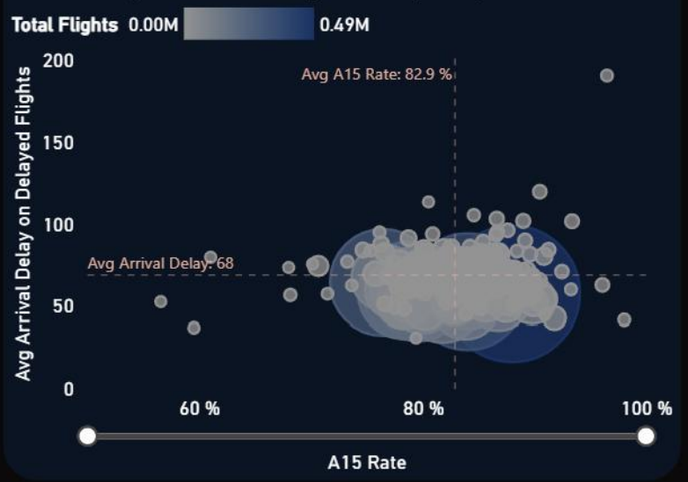
Hover for more information

Total Flights ● A15 Flights

A15 Flights | A15 Rate



A15 & Avg Arrival Delay Rate by Airport



Total Arrival Delay (min) by Airport & Reasons

Hover for more information

Air System ● Weather ● Security



Filter

- Month Name: All
- AIRPORT: All
- AIRLINE: All

Canceled & Diverted Flights by Airport

Hover for more information

Cancelled Flights ● Diverted Flights



Avg Taxi In VS Avg Arrival Delay

Switch to view

Taxi Out | Taxi In



Average Departure Delay (min) by time of a day

AIRPORT	00–04	05–07	08–10	11–13	14–16	17–19	20–23
Aberdeen Regional Airport		266		107	56		
Abilene Regional Airport		116	94	53	53	57	23
Abraham Lincoln Capital Airport		166	84	77	68	81	
Adak Airport						36	
Akron-Canton Regional Airport		106	75	69	72	60	197
Albany International Airport		96	70	57	60	62	45

Route Map

AIRLINE

Alaska Airlines Inc.	American Airlines Inc.	American Eagle Airlines Inc.	Atlantic Southeast Airlines	Delta Air Lines Inc.	Frontier Airlines Inc.	Hawaiian Airlines Inc.	JetBlue Airways	Skywest Airlines Inc.	Southwest Airlines Co.	Spirit Air Lines	United Air Lines Inc.	US Airways Inc.	Virgin America

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ORIGIN AIRPORT

Search

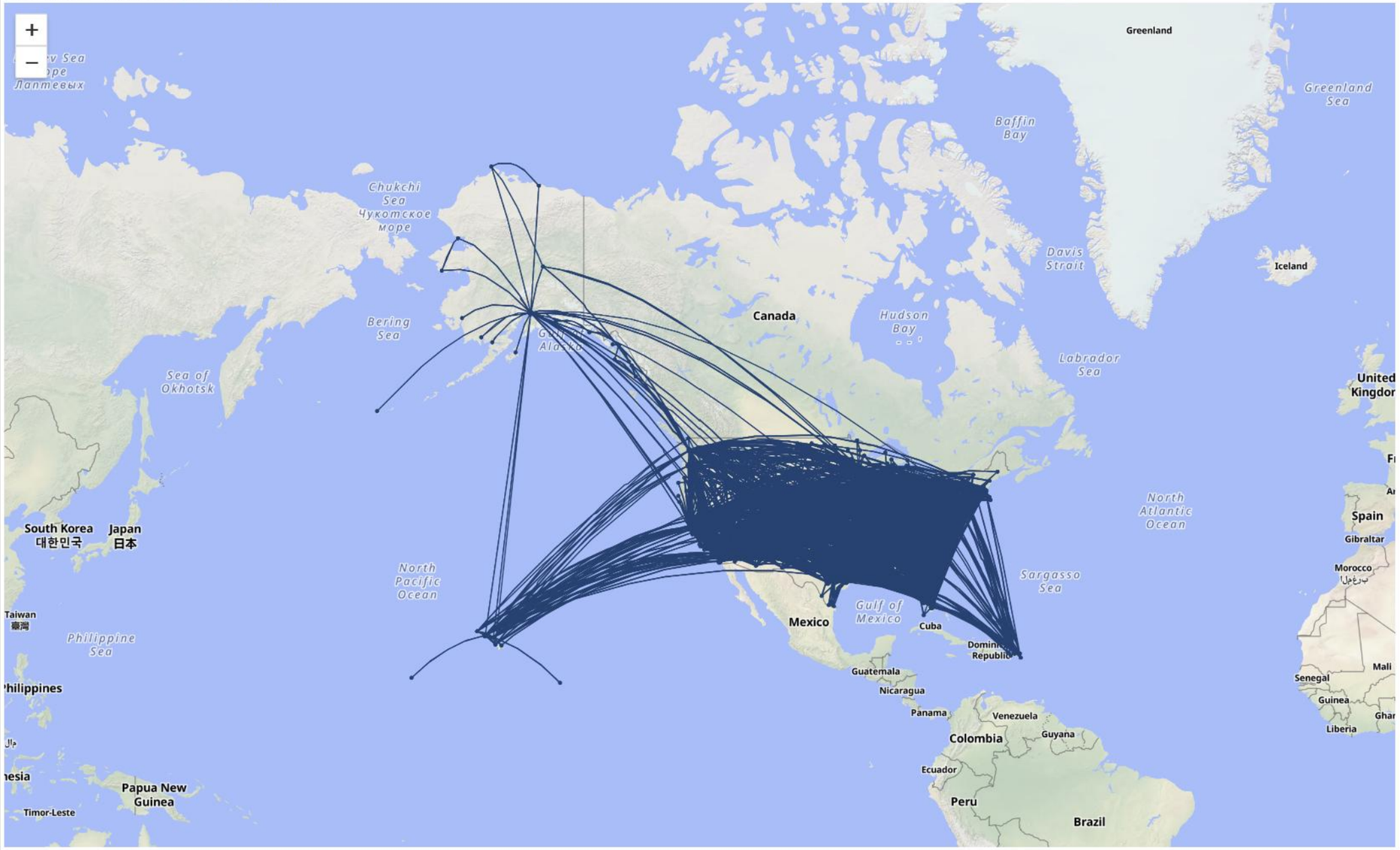
- Select all
- Aberdeen Regional Air...
- Abilene Regional Airp...
- Abraham Lincoln Capi...

DESTINATION AIRPORT

Search

- Select all
- Aberdeen Regional Air...
- Abilene Regional Airp...
- Abraham Lincoln Capi...

Line Width = Total Arrival Delay (min)



ROUTE

Search

- Select all
- ABE -> ATL
- ABE -> DTW
- ABE -> ORD

CITY

Search

- Select all
- Aberdeen
- Abilene
- Adak